

Divergences: robust objectives and the KL limit I

The generalized-Bayes objective eq. 1 has exactly two tunable ingredients: the divergence D that regularizes the update toward the prior or cavity, and the loss L that measures data fidelity. This section develops both — the divergence family first, the robust loss family in sec. `sec:method-losses` — and shows that each carries a limit in which it collapses to its standard counterpart, KL for the divergence and NLL for the loss. Those client-side limits establish recovery to standard Bayes. The distinct categorical server bridge in sec. 5.8 then identifies a qualified log-linear message-combination specialization; it does not recover the complete source belief-sharing protocol.

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The regularizing divergence D decides how far a client's updated belief may move from its cavity, so choosing D is a modeling decision rather than a numerical detail. The family lives in `divergences.py`. We implement the forward KL (the standard-Bayes case), the reverse KL (FedGVI's RKL client divergence), the standard α -Rényi diagnostic, FedGVI's Alpha-Rényi normalization (AR), and total variation (a bounded distance in $[0, 1]$). The single most important recovery property is that the robust members recover the KL divergence in a limit:

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$$D_{\alpha}(q \parallel p) \xrightarrow{\alpha \rightarrow 1} \text{KL}(q \parallel p). \quad (1)$$

Lemma (KL is the $\alpha \rightarrow 1$ limit of the Rényi family)

For categorical pmfs q, p on a finite support, the α -Rényi divergence $D_{\alpha}(q \parallel p) = (\alpha - 1)^{-1} \log \sum_k q_k^{\alpha} p_k^{1-\alpha}$ tends to $\text{KL}(q \parallel p)$ as $\alpha \rightarrow 1$, the limit (1). The `divergences.py` implementation switches to the KL closed form inside a small band around $\alpha = 1$, so on that band the equality is exact rather than merely asymptotic.

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KL is the divergence that makes generalized Bayes collapse to standard Bayes. When local posteriors are then combined by the separately specified categorical message-combination specialization in sec. 5.8, the project recovers its log-linear-pool corner; neither step reconstructs the complete belief-sharing protocol of Friston et al. (Friston et al. 2024). Everything robust is a controlled departure from that fixed point; Lemma 1 is the formal hinge, and the largest observed Rényi-versus-KL discrepancy in the recovery band is 0 (reported in sec. 7.1).

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The standard Rényi diagnostic is `renyi_divergence`; FedGVI's AR regularizer is `alpha_renyi_divergence`, equal to the standard form divided by α . For the finite categorical support, `generalized_posterior` solves the named Alpha-Rényi objective through its scalar normalization condition rather than using a generic power-softmax shortcut. This distinction keeps the reported limit and the implemented objective aligned. The AR regularizer is not merely a diagnostic: it is exercised as a client divergence in the categorical FedGVI baseline of sec. 7.6, where it pairs with the `rcce` loss of sec. `sec:method-losses` to constitute the genuine per-client robustness axis.

Robust losses: bounded influence at the Bayes corner I

The data-fidelity term of eq. 1 lives in `losses.py`. Standard Bayes uses the negative log-likelihood, $\text{NLL}(p, o) = -\log p(o)$, which is *unbounded*: a single contaminated observation with $p(o) \rightarrow 0$ dominates the posterior. This is precisely the fragility the robust-Bayes literature was built to remove (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015; Zhang and Sabuncu 2018), extended into robust-divergence variational inference (Futami et al. 2018), and the property FedGVI imports into federated inference (Mildner et al. 2025). The robust-statistics vocabulary here is the usual influence-function one (Huber and Ronchetti 2009): bounded losses reduce the leverage of extreme observations, while NLL does not. We implement two categorical robust losses, each of which recovers NLL in a limit.

Robust losses: bounded influence at the Bayes corner I

The density-power (β) loss (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015; Futami et al. 2018) is recentered so that the scalar limit is exact:

$$L_{\beta}(p, o) = -\frac{p(o)^{\beta} - 1}{\beta} + \frac{\sum_k p_k^{\beta+1} - 1}{\beta + 1}, \quad L_{\beta} \xrightarrow{\beta \rightarrow 0} \text{NLL}. \quad (2)$$

The robust categorical cross-entropy (generalized cross-entropy) (Zhang and Sabuncu 2018) is

$$L_{q_{\text{loss}}}(p, o) = \frac{1 - p(o)^{q_{\text{loss}}}}{q_{\text{loss}}}, \quad L_{q_{\text{loss}}} \xrightarrow{q_{\text{loss}} \rightarrow 0} \text{NLL}, \quad (3)$$

Robust losses: bounded influence at the Bayes corner I

which by l'Hôpital recovers NLL as $q_{\text{loss}} \rightarrow 0$ and at $q_{\text{loss}} = 1$ is the bounded mean-absolute-error loss $1 - p(o)$, finite exactly where NLL diverges.

Proposition (β -loss and rcce recover NLL)

The recentered density-power loss L_β of (2) tends to the negative log-likelihood as $\beta \rightarrow 0$, and the robust categorical cross-entropy $L_{q_{\text{loss}}}$ of (3) tends to the negative log-likelihood as $q_{\text{loss}} \rightarrow 0$. Both limits are exact in the implementation; the largest observed $\beta \rightarrow 0$ discrepancy from the NLL closed form is 0 and the largest $q_{\text{loss}} \rightarrow 0$ discrepancy is 0 (Section 7.1). At the bounded end the loss stays finite where NLL diverges, the source of the robustness validated in Section 7.5.

Robust losses: bounded influence at the Bayes corner I

Taking the loss-parameter limits ($\beta \rightarrow 0$ or $q_{\text{loss}} \rightarrow 0$) reproduces standard Bayes. Combining those local posteriors through the qualified categorical specialization in sec. 5.8 is a separate server step, not a recovery claim for the complete belief-sharing protocol of Friston et al. (2024). This is the **per-agent rigorous robustness axis**: it is derived from eq. 1 and provably limits to Bayes through Proposition 1 and Lemma 1, and it is the axis that carries FedGVI's bounded-influence guarantee under the cited matching assumptions. The complementary server-side divergence-reweighting heuristic of sec. 5.8 is a distinct device and is never granted this guarantee (sec. 3.3).